

MTH102 – Probability and Statistics
(1st Mid-semester examination)

5th of February, 2026

Duration: 1 hour.

You **MUST** give **complete explanations** for all your answers.

Name: _____ Id: _____

1. A biased coin, which lands on head with probability 0.9, is tossed 10 times. Assume that the tosses are independent events.
 - (a) (1 point) What is the probability of getting all heads?
 - (b) (1 point) What is the probability of getting at least 3 tails?
 - (c) (1 point) What is the probability of getting a head in the first and last toss?
2. A box has 10 white balls and 6 black balls. 7 balls are chosen at random.
 - (a) (1 point) What is the probability that all of them are white?
 - (b) (1 point) What is the probability that all of them are black?
 - (c) (1 point) What is the probability that at least 3 of them are black?
3. (3 points) Consider an event A . If the conditional probability that A occurs given that B occurs is the same as the conditional probability that A occurs even when B does **not** occur, then show that A and B are independent events, i.e. show that $P(A \cap B) = P(A)$.
4. A pair of dice are rolled 3 times.
 - (a) (3 points) What is the probability of getting a sum of 12 in at least one of these three rolls?
 - (b) (2 points) What is the probability of getting a sum of 1 in the first roll?
5. (3 points) There are 10 coins in a box, 9 of which are unbiased, and 1 biased one. The biased coin lands on a head with probability 0.8. One of the coins is chosen at random and tossed 10 times. Each of the 10 times, the coin lands on a head. What is the probability that the biased coin was picked?
6. (3 points) A box has many balls, each numbered either 1, 2, or 3. The probability of finding a ball numbered 2 is twice the probability of finding a ball numbered 1, and the probability of finding a ball numbered 3 is twice the probability of finding a ball numbered 2. What is the probability of finding a ball with a number less than or equal to 2?